

Total No. of Questions : 8]

SEAT No. :

PE-4258

[Total No. of Pages : 2

[6582]-30

S.E. (Electronics & Computer Engineering)
PRINCIPLES OF PROGRAMMING LANGUAGE
(2019 Pattern) (Semester - IV) (204206)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary.

Q1) a) Explain functional and logic programming paradigms. **[5]**

b) Explain in detail about **[6]**

i) Abstraction.

ii) Modularity.

c) Explain Interface in Java with examples. **[6]**

OR

Q2) a) What is meant by modularity? Which concepts are supported to achieve modularity in any programming language? **[6]**

b) Explain the concept of Encapsulation in detail with example. **[6]**

c) What is Generic data structures? Explain with examples. **[5]**

Q3) a) Explain Java Virtual Machine. **[6]**

b) Difference between Users defined and Primitive data types. **[6]**

c) Explain any Java Iterative statement with example **[6]**

i) Loop statement

ii) Jump statement

OR

P.T.O.

- Q4)** a) Explain architecture of Java virtual machine. [6]
b) Explain the concept of any two arrays. How to declare an array in Java? Compare arrays in C and Java. [6]
c) Explain data types and sign in Java with examples. [6]

- Q5)** a) What is difference between method overriding and method overloading. [5]
b) Explain various forms of implementing an interface in Java. How multiple inheritance is supported in Java? [6]
c) What is use of package in Java? How CLASSPATH helps to store and retrieve the classes? How is access protection provided for packages? [6]

OR

- Q6)** a) What is an interface in Java? How is this different than a class? Give example of interface. [6]
b) Explain concept of method overloading in Java with suitable example. [5]
c) What is method overriding? In Java, what is advantage of using overriding? Demonstrate method overriding with example. [6]

- Q7)** a) What are uncaught exceptions? State the use of try(), catch(), throw() methods. [6]
b) Explain the method of file input stream and file output stream classes in Java. [6]
c) What are stream classes in Java? List and explain the methods of Character Array output Stream class. [6]

OR

- Q8)** a) Write a program for use of multiple catch blocks in Java. [6]
b) State the use of following methods for programming applet. Give an example of using each of these init(), Start(), Paint(), Stop(), Destroy(), Update(). [6]
c) Explain in detail about applet architecture. [6]

